

NeuroPix: AI-Enhanced Image Processing and Management Web Application

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ABSTRACT

Many image-editing tasks require users to move a file between a standard editor, an AI service, and separate storage. NeuroPix brings these steps into an authenticated web application.

Pillow handles standard adjustments, while OpenAI and a locally hosted Stable Diffusion 1.5 model process AI edits. Flask connects the interface to MySQL and Amazon S3, which store account details, image records, and files. Users can work from the same uploaded image, compare each result with the original, and keep completed edits in a personal gallery.

INTRODUCTION

Combining conventional edits with AI changes usually means moving an image between tools: crop in one editor, upload to an AI service, compare the result, then organize saved files by hand. An open-ended AI prompt also forces users to describe both the change and what must stay the same.

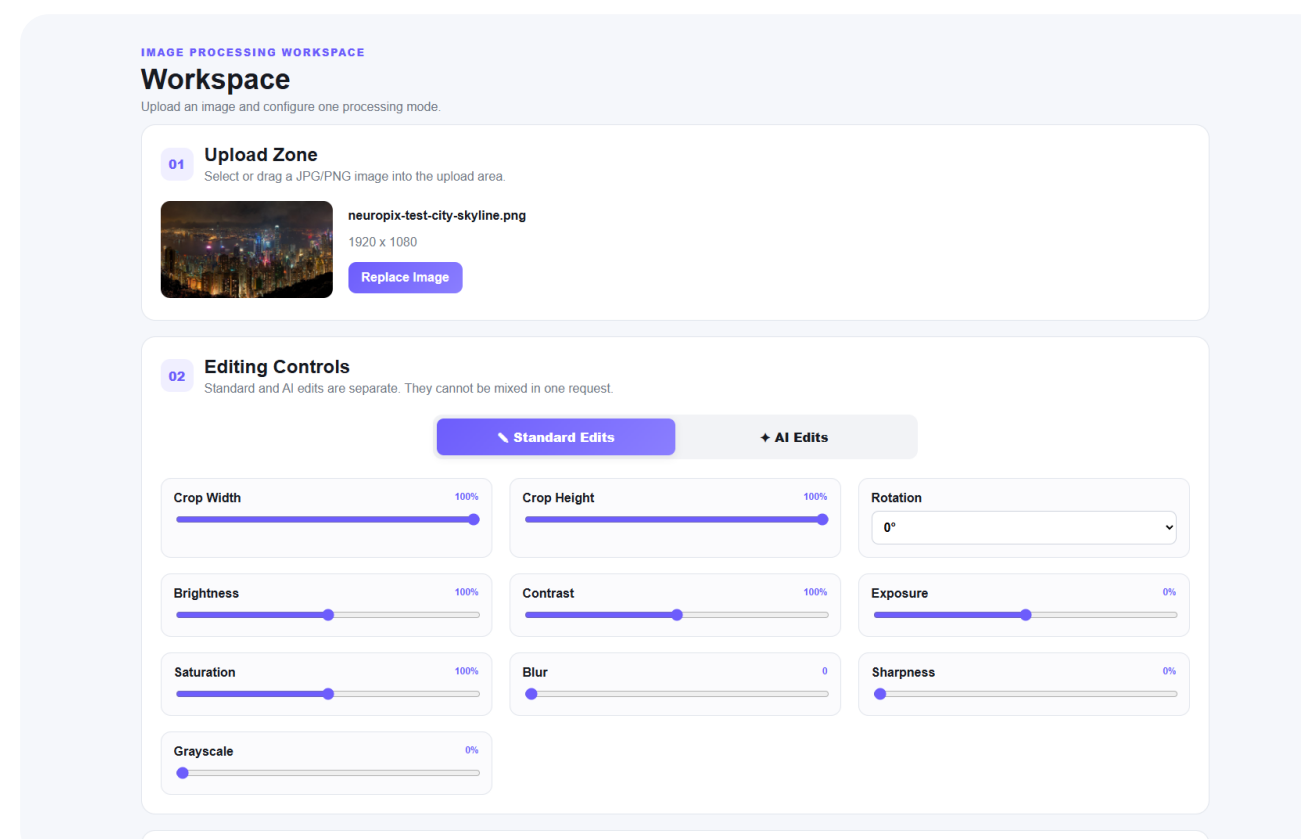
NeuroPix keeps the source, editing controls, comparison, and saved results together for users with basic browser skills.

Objectives

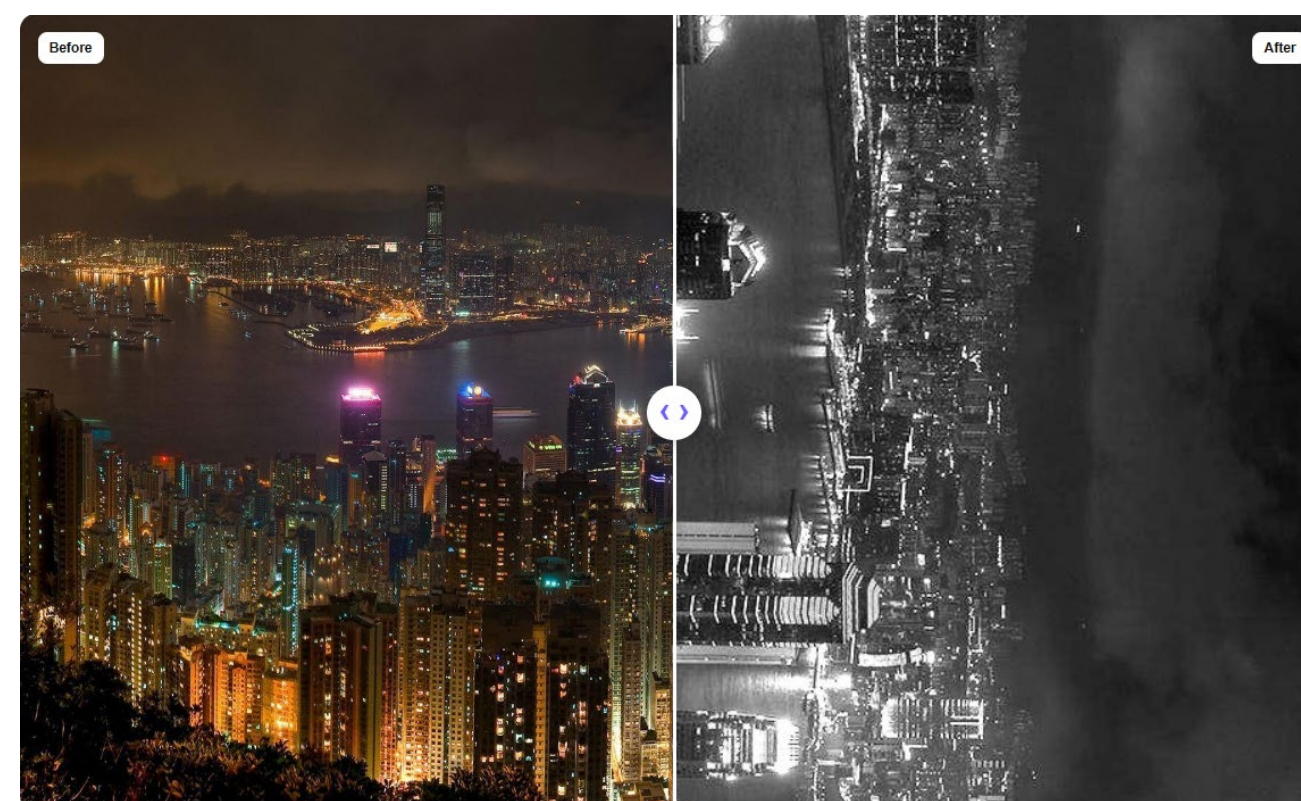
- Browser-based JPG/PNG editing with no desktop editor
- Standard controls plus purpose-specific AI fields
- Images in Amazon S3, metadata in MySQL
- Comparison, download, and a personal gallery

RESULTS

NeuroPix works as one workspace: the image is uploaded once in the Studio and stays in place while the user edits it. Standard mode covers everyday adjustments through sliders, and AI mode takes a short description when a change is harder to make by hand. Every result is saved to the account, so it can be compared with the original, downloaded, or reopened later.



The Studio: upload an image and set the controls



A Standard edit in the comparison view



The gallery of saved results

SUMMARY

NeuroPix was created for users who want standard image controls and AI editing without moving files between several applications. After signing in, a user can upload an image, apply familiar adjustments or select an AI provider, and compare the result with the original.

Guided fields separate modification, background, and enhancement requests, so the user does not need to write one detailed prompt. Completed images can be downloaded, reopened for further editing, or removed from a private gallery. NeuroPix brings the complete editing process into one connected browser workspace.

CONCLUSIONS

NeuroPix: a deployed web application for Standard and AI-assisted editing, comparison, download, and personal image management, built within the team's budget and hardware.

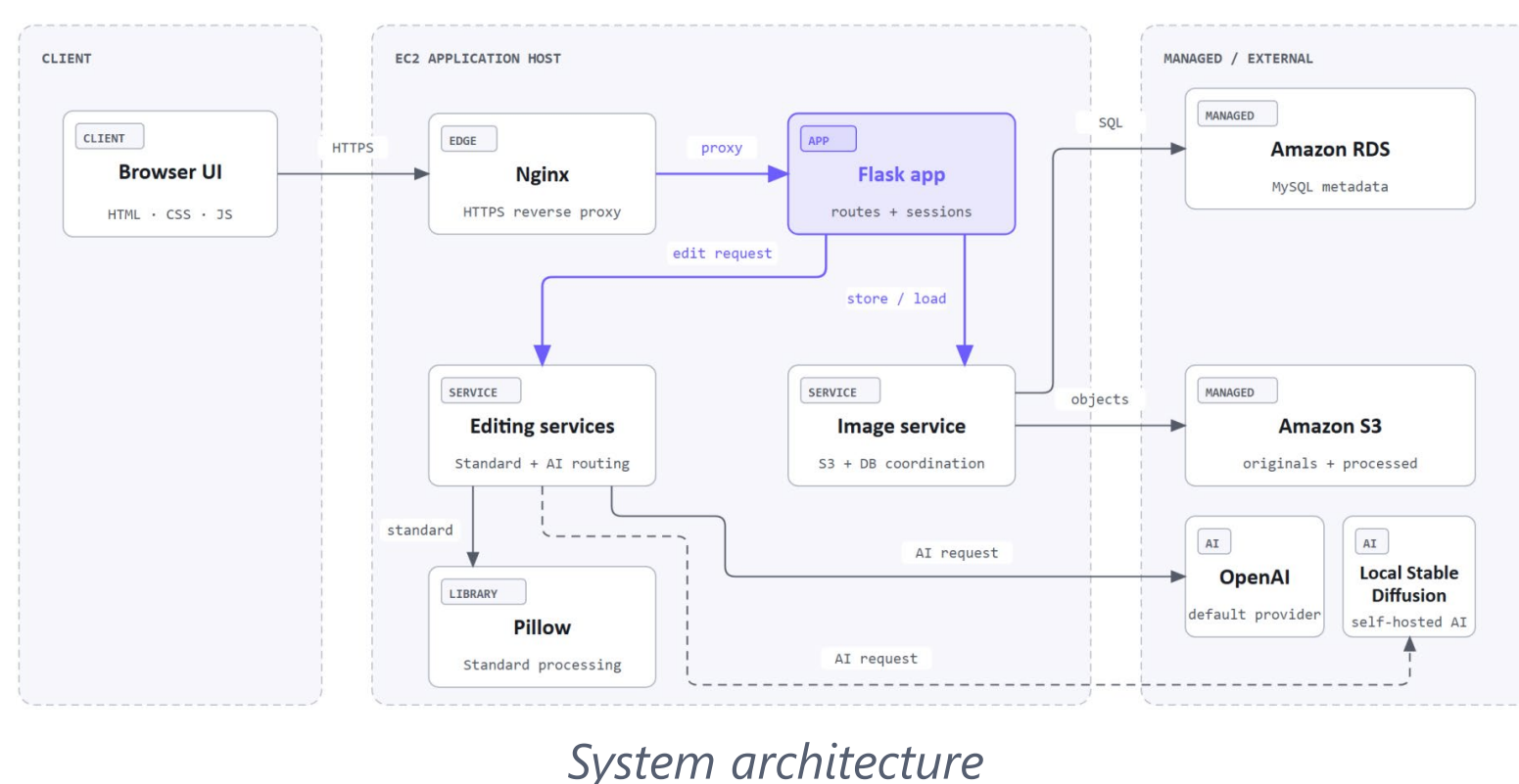
Building it taught us where the trade-offs sit. Waiting for an AI result was the part users feel most, and running our own model kept costs down while tying the service to one computer. Designing around one clear path through the site mattered more than adding features.

Future work

- Move AI work to a background queue with progress reporting
- Improve recovery when a save is incomplete
- Adopt feature branches, code review, and staging
- Better local output quality within RTX 3080 limits

METHODS

- Flask backend with an HTML/CSS/JS frontend
- Standard mode: ten Pillow controls (crop, rotation, contrast, blur, ...) in a fixed order
- Structured pre-prompting: modification, background, and enhancement fields, with scope and preservation context added before calling OpenAI
- Local AI: Stable Diffusion 1.5 in Docker on an RTX 3080 via a reverse SSH tunnel
- Hosted on AWS EC2 with Nginx, HTTPS, RDS, and S3



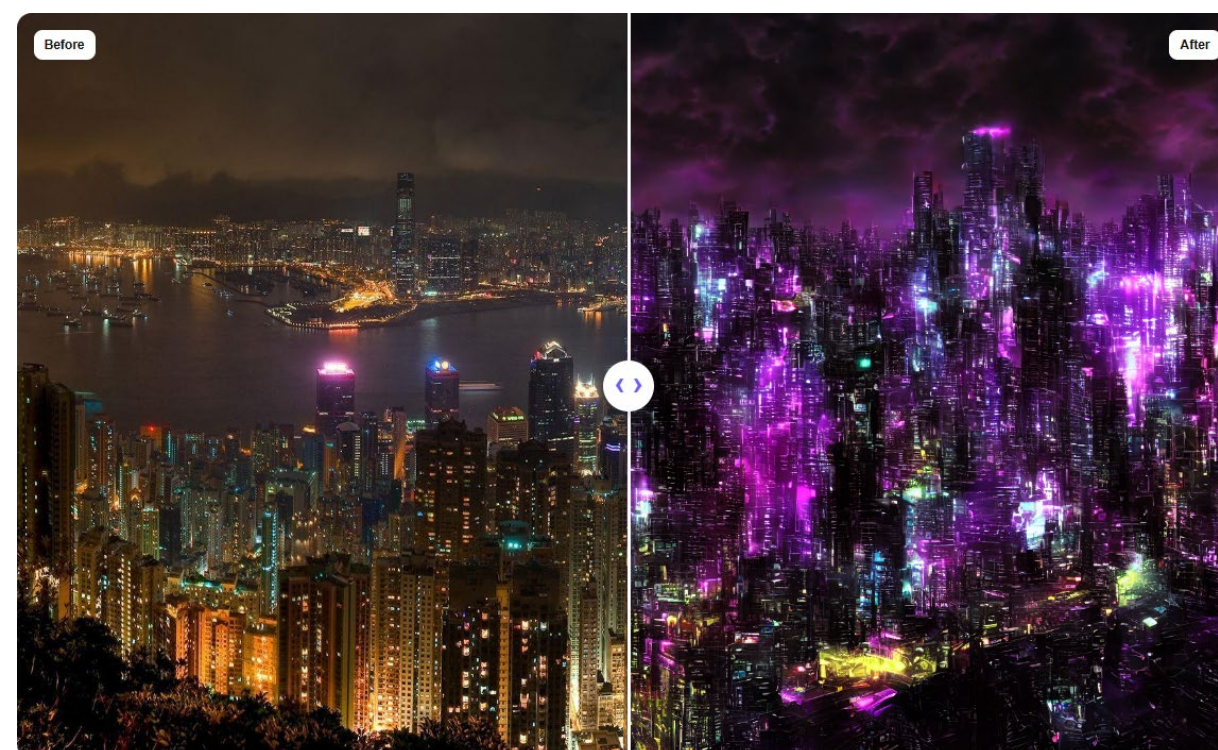
System architecture

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AI edit made with OpenAI



AI edit made with local model

Both AI paths were given the same photo and the same prompt. OpenAI returned a neon night city with rain and a large moon while keeping the skyline composition, and the self-hosted model returned a stronger purple, futuristic version of the same scene.